Exercise Sheet 1

Due: 27.10.26 9h00

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General Regulations.

- Exercises are not graded. Submission is optional; tutors provide feedback.
- Work in groups of two or three, preferably from the same tutorial group. Submit all files via the Übungsgruppenverwaltung, once per group, and include the names of all group members.
- Theoretical solutions may be handwritten and scanned or typeset using LaTeX. Please ensure that scans are legible and not blurry.
- Course material is available in the GitHub repository.

1 Vector calculus in numerator layout

Solve this exercise yourself, without an AI agent.

All vectors are columns. For a differentiable map $\mathbf{y} : \mathbb{R}^n \to \mathbb{R}^m$, use the lecture's *numerator layout*:

$$\left[\frac{\partial \mathbf{y}}{\partial \mathbf{x}}\right]_{ij} = \frac{\partial y_i}{\partial x_j}, \qquad \frac{\partial \mathbf{y}}{\partial \mathbf{x}} \in \mathbb{R}^{m \times n}.$$

For a scalar f, the derivative $\partial f / \partial \mathbf{x}$ is a row; its transpose is the column gradient $\nabla_{\mathbf{x}} f = (\partial f / \partial \mathbf{x})^\mathsf{T}$. Derive the results either by writing the equations with explicit indices and summations or by reusing statements you have already proven. State the dimensions of each derivative and gradient.

(a) Linear maps. For constant $A \in \mathbb{R}^{m \times n}$ and $\mathbf{c} \in \mathbb{R}^n$, show that

$$\frac{\partial (A\mathbf{x})}{\partial \mathbf{x}} = A, \qquad \frac{\partial (\mathbf{c}^\mathsf{T}\mathbf{x})}{\partial \mathbf{x}} = \mathbf{c}^\mathsf{T}.$$

Give the gradient of $\mathbf{c}^\mathsf{T}\mathbf{x}$.

(b) Product rule. For differentiable maps $\mathbf{a}, \mathbf{b} : \mathbb{R}^n \to \mathbb{R}^k$, prove

$$\frac{\partial (\mathbf{a}^\mathsf{T}\mathbf{b})}{\partial \mathbf{x}} = \mathbf{a}^\mathsf{T} \frac{\partial \mathbf{b}}{\partial \mathbf{x}} + \mathbf{b}^\mathsf{T} \frac{\partial \mathbf{a}}{\partial \mathbf{x}}.$$

Annotate the dimensions of the factors to check that both products give row derivatives.

(c) Quadratic forms. Let $B \in \mathbb{R}^{n \times n}$ be constant, without assuming symmetry. Use parts (a) and (b) to derive the row derivative and column gradient of

$$q(\mathbf{x}) = \mathbf{x}^\mathsf{T} B \mathbf{x}.$$

Simplify your results when $B = B^\mathsf{T}$ and when $B = I$.

(d) Chain rule and a squared residual. For differentiable maps $\mathbf{g} : \mathbb{R}^n \to \mathbb{R}^m$ and $\mathbf{h} : \mathbb{R}^m \to \mathbb{R}^r$, derive

$$\frac{\partial\, \mathbf{h}(\mathbf{g}(\mathbf{x}))}{\partial \mathbf{x}} = \left.\frac{\partial \mathbf{h}}{\partial \mathbf{y}}\right|_{\mathbf{y}=\mathbf{g}(\mathbf{x})} \frac{\partial \mathbf{g}}{\partial \mathbf{x}}.$$

Apply this rule to compute the row derivative and column gradient of the least-squares objective

$$L(\mathbf{x}) = \|A\mathbf{x} - \mathbf{d}\|_2^2, \qquad A \in \mathbb{R}^{m \times n}, \quad \mathbf{d} \in \mathbb{R}^m,$$

with A and $\mathbf{d}$ constant. Check your result by expanding L and using parts (a) and (c).

2 PCA as non-convex optimization problem

Use the agent only to write the required code for this exercise. Answer all other questions yourself. Do not hand in the web app; submit only your answers to the questions.

A very useful application for coding agents is to build interactive visualizations for a given topic. In this exercise we will build an interactive web app to demonstrate various properties of PCA. We recommend using HTML, CSS, and JavaScript, with calculations and visualizations running directly in the browser.

Draw observations from a zero-mean Gaussian with covariance

$$\Sigma_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \qquad \rho \in (-1, 1),$$

starting with $n = 300$ and $\rho = 1/2$. Subtract the sample mean and collect the centered observations in $X \in \mathbb{R}^{2\times n}$. Write

$$C = \frac{1}{n} X X^\mathsf{T}, \qquad V(\mathbf{w}) = \mathbf{w}^\mathsf{T} C \mathbf{w}.$$

Here V is the projected variance for unit $\mathbf{w}$. Let $\lambda_1 \geq \lambda_2$ be the eigenvalues of C, with $\lambda_1 > 0$, and let $\mathbf{w}_*$ be a unit PC1 vector.

(a) Build an interactive PCA web app. Ask the agent to build a web app containing two linked plots with equal axis aspect ratios:

- The data cloud along with the optimal PC1 line.
- Contours of $-V(\mathbf{w})$, with the unit circle and the optimal directions $\pm\mathbf{w}_*$ marked.

Include controls for the sample count, independent scaling of both features and correlation. Keep the underlying random draws fixed while adjusting any of these controls, and center each resulting dataset. Do not standardize after scaling.

(b) Explore parameter effects. How do the loss landscape and PCA solutions change when you scale one feature, scale both equally, change the correlation, or change the sample count? Make predictions before using the controls and explain your observations using C.

(c) Solve the constrained optimization iteratively. PCA solves

$$\max_{\mathbf{w}\in\mathbb{R}^2} V(\mathbf{w}) \quad \text{subject to} \quad \mathbf{w}^\mathsf{T}\mathbf{w} = 1.$$

The objective is convex (see part (d)), whereas the constraint set is non-convex. We can seek a solution iteratively by gradient ascent followed by normalization:

$$\mathbf{u}_k = \mathbf{w}_k + 2\eta C \mathbf{w}_k, \qquad \mathbf{w}_{k+1} = \frac{\mathbf{u}_k}{\|\mathbf{u}_k\|_2}, \qquad \eta = \frac{1}{2\,\mathrm{tr}\,C}.$$

Ask the agent to extend your app with an initial-angle control, an iteration slider, and play/pause animation. Show the trajectory of $\mathbf{w}_k$ on the loss landscape as well as the corresponding PC line in the data plot. Keep the dataset and step size fixed during each run. Does this iteration converge to a PCA optimum from every unit initialization? Can you reproduce your answer in your app?

(d) Relax the constraint. Prove that V is convex and that replacing $\mathbf{w}^\mathsf{T}\mathbf{w} = 1$ by the convex constraint $\mathbf{w}^\mathsf{T}\mathbf{w} \leq 1$ preserves both the maximum and the maximizers when $\lambda_1 > 0$. You may use $V(r\mathbf{q})$ for a unit vector $\mathbf{q}$. Does the relaxed problem qualify as a convex optimization problem?

(e) Compare optimization on the circle and the disk. Ask the agent to extend the same app to allow initializations throughout the unit disk. Implement gradient ascent followed by Euclidean projection onto the disk:

$$\mathbf{v}_{k+1} = \frac{\mathbf{v}_k + 2\eta C \mathbf{v}_k}{\max\{1, \|\mathbf{v}_k + 2\eta C \mathbf{v}_k\|_2\}}.$$

Use the same dataset and step size for both methods and add linked playback. For the initial Gaussian sample, take a shared unit direction $\mathbf{q}$ and compare $\mathbf{w}_0 = \mathbf{q}$ with $\mathbf{v}_0 = r_0\mathbf{q}$, where $0 < r_0 < 1$. Assume C is positive definite for this comparison. Overlay the trajectories in parameter space and compare them. What do you observe as the iteration progresses? Can you explain theoretically whether and when the full iterates become identical? What happens for $\mathbf{v}_0 = \mathbf{0}$?

3 RANSAC

Solve this exercise yourself, without an AI agent.

Consider a dataset in which a fraction $p \in (0, 1)$ of the points are sampled from the inlier distribution, while the remaining points are outliers. You want to find a good initialization for a robust estimator. You repeatedly and independently sample m points, in the hope of obtaining at least one sample which is free from contamination. How many repeats r are needed to find at least one outlier-free sample with a probability of at least 99%? Assume that the points within each sample are drawn with replacement, i.e. a sample may contain duplicates.

Hint: For a single sample of m points, what is the probability of it being outlier-free?